Painter Names and Feature Distributions: Comparing Generated Images with Original Paintings

Anonymous authors
Research manuscript prototype

7 September 2026

Abstract

An image can suggest a painter's style while occupying a different distribution of measurable image properties. We investigate this distinction using 31 fixed, interpretable color, spatial and digital-texture features. Completed development analyses compare 649 digital painting surrogates by Monet, Sisley, Pissarro and Cézanne with 1,920 generated images from two requested GPT Image service aliases, three prompt methods and matched artist-free controls. Across painter-conditioned cells, generated/original total sample-variance ratios range from 0.206 to 0.376, and scene/work-disjoint RBF classification reaches balanced accuracy 0.940–0.983. Artist-free images are also readily distinguished (0.915–0.990), and broad content reweighting changes the spread ratios little. These observations support an empirical feature-distribution gap, while leaving its painter specificity, perceptual meaning and capture-related causes unresolved. We therefore design a focused extension with two additional model families, visually coded reference content and a prompt-specificity intervention. An 18-image technical pilot succeeds at a total paid cost of $0.8305, but reveals encoding differences between the paid routes and render deviations on the OAuth comparator. This prototype reports the completed evidence and distinguishes it from the ongoing controlled extension; it does not claim completed new-model fidelity results or human validation.

1 Introduction

The question motivating this study is whether painter-conditioned image generation reproduces the distribution of visual properties found across a painter's original works. A small distance to a painter's average feature vector does not answer this question: generated images can cluster tightly near that average while failing to reproduce variation across works. Conversely, a recognizable artistic reference may coexist with systematic differences in palette, spatial organization or digital texture.

We distinguish four properties: distribution discrepancy, within-set spread, reference coverage and painter specificity. None is an aesthetic quality score. The measured objects are digital reproductions of paintings and generated image files. Their comparison combines artistic choices with content, reproduction, resizing, encoding and service behavior. Treating every detectable difference as a failure of painter-style reproduction would overstate what the measurements identify.

The completed analysis suggests a useful, narrower hypothesis: under the tested prompt libraries and image processing, painter-conditioned generations occupy more concentrated and distinguishable feature distributions than the recorded original-painting surrogates. The extension asks whether this pattern persists under stronger content controls and across additional model families. Its contribution is intended to be explanatory evidence about the gap, rather than another demonstration that two scatter clouds can separate.

2 Related work and the distinct research question

Quantitative art research has characterized painting collections using color, image statistics and information-theoretic measures (Kim et al., 2014; Lee et al., 2020; Sigaki et al., 2018). Kim et al.'s 2026 study concerns contextual multimodal navigation through art-historical development (Kim et al., 2026). Our sampling unit and target differ: we compare within-painter distributions across distinct works with newly generated text-to-image collections, with painter-name and prompt-content interventions. We do not claim to replicate its art-evolution analysis or its complete measurement system. Our fixed 31-feature representation is the existing implementation in this project.

Asperti's recent preprint reports separation of human and generated paintings in CLIP space and investigates preprocessing robustness, interpretable multiscale structure and inversion (Asperti, 2026). Thus, scatter separation alone is not sufficient novelty for this work. We emphasize painter-conditional comparisons, explicit artist-free controls and the effects of prompt content on a fixed interpretable representation. We have not run CLIP or reproduced that preprint's inversion experiments. More generally, feature-space distinctions need not have the perceptual significance assumed by a style evaluator.

Generative-model evaluation separates fidelity and coverage for related reasons (Naeem et al., 2020). Resizing and compression can change measured differences (Parmar et al., 2022); digital-surrogate provenance is consequently part of the experimental design rather than a minor file-handling detail. Energy statistics provide a distributional comparison based on pairwise distances (Székely and Rizzo, 2013), without reducing each collection to its centroid.

3 Completed development data and methods

3.1 Reference paintings and generation accounting

The reference analysis uses 649 measured original-painting surrogates: 297 Monet, 106 Sisley, 141 Pissarro and 105 Cézanne. The broader recorded frame also contains development and qualification works; they cannot be relabeled as new holdouts. A separate 221-work development set supplies the frozen feature scaler.

The repeated prompt study has two requested service aliases, `gpt-image-1` and `gpt-image-2`; three methods (by name, explicit style instruction, and style instruction plus aspects); 16 scene templates; and four repetitions. For each alias/method there are four named painter conditions and an artist-free condition. This gives 1,920 planned slots, with 64 images per condition. The original run returned 1,918 images and two moderation refusals. Two later, explicitly authorized requests used the exact refused payloads and succeeded. The derived descriptive grid therefore contains 1,920 measured images from 1,922 requests: 1,536 named and 384 artist-free images.

The two later images do not restore their original randomized time positions. The original complete-grid primary remains unavailable, while its source refusals and earlier analyses remain preserved. The descriptive results below use the completed derived grid. We do not attach new randomization p-values or confidence intervals to that retrospective replacement. Requested aliases are service labels; the available evidence does not attest distinct underlying model snapshots. Accordingly, they are not treated as two independently verified model families.

3.2 Feature representation and processing

Each image is represented by 11 color, eight spatial and 12 digital-texture features. Color includes lightness and chroma location/spread, hue organization, and multiscale chromatic differences. Spatial features include spectral shape, orientation organization, gradients and quadrant differences. Texture includes wavelet energies and trends, local binary-pattern entropies and local luminance variation. These are measurements of digital images, not direct measurements of physical brushstroke relief or a validated complete definition of painter style.

The primary normalization applies EXIF orientation, converts valid embedded profiles to sRGB, assumes sRGB when compatible unprofiled data are present, rejects nonopaque images and preserves aspect ratio at a 512-pixel short side. It uses Lanczos resampling without upsampling or discretionary cropping. Each coordinate is centered by the development median and divided by the development interquartile range. Development painters receive equal total mass; no generated or new reference feature is used to fit that scaler.

3.3 Distribution discrepancy, spread and content weights

Let x_i and y_j be standardized reference and generated vectors, with nonnegative weights a_i and b_j that each sum to one. We report the empirical V-statistic energy discrepancy

$$\widehat{\mathcal{E}}(X,Y) = 2\sum_{i,j} a_i b_j \|x_i - y_j\|_2 - \sum_{i,k} a_i a_k \|x_i - x_k\|_2 - \sum_{j,l} b_j b_l \|y_j - y_l\|_2. \tag{1}$$

All pairwise terms contribute; the reference mean is not the sole comparison target. The finite-sample V-statistic has a nonzero sampling baseline even when independent samples come from one common population.

We report the ratio of total within-group sample variances for ordinary unweighted data. Weighted analyses use the population-form trace $\sum_i a_i \|x_i - \sum_k a_k x_k\|^2$ in both numerator and denominator. The sample and population definitions have different finite-sample denominators. A complementary robust summary divides the sum of squared generated coordinate IQRs by the corresponding reference sum. Undefined coordinate ratios remain undefined rather than being stabilized by an arbitrary constant.

Content weighting assigns equal total mass to four previously recorded broad classes. Original labels come from title metadata; generated labels come from the intended prompt. This is a sensitivity to the assigned content mixture, not independent validation of actual depicted content. The old title lexicon can confuse geographic names with composition, and the four classes cannot control all subjects, viewpoints or scene complexity.

3.4 Real/real baselines and separation

For each painter, 128 deterministic draws partition distinct original works into two disjoint groups of size $n = \min(64, \lfloor N/2 \rfloor)$. Matched-$n$ generated/original comparisons use the same original subsets. These repeated draws reuse one corpus; they are not 128 independent studies, population confidence intervals or an artistic-equivalence margin.

Common PCA views retain all points and shared axes within a painter. PCA is descriptive and is not used to train the detector. Fixed linear and RBF kernel-ridge classifiers use ridge 0.01, class-balanced training and no hyperparameter search. Test folds hold out complete generated scene families and distinct original works. Transfer analyses additionally hold out the target painter or requested alias as specified in the released membership records. Group-disjoint evaluation addresses leakage from repeated prompts and works (Roberts et al., 2017); it does not establish independent backend sessions.

4 Completed development results

4.1 Concentration and separability

Table 1 summarizes the all-31-feature cells. Named generation has substantially smaller empirical spread than the originals under both variance and IQR-based summaries. The artist-free cells have larger spread ratios than the named ranges but are also distinguishable from originals.

Condition	Sample variance	Content-equal variance	IQR spread	RBF BA
Named	.206–.376	.211–.372	.215–.396	.940–.983
Artist-free	.369–.750	.378–.728	.352–.646	.915–.990

Table 1: Ranges over four painters, two requested aliases and three prompt methods. Spread values are generated/reference ratios. BA is balanced accuracy. The content-equal column uses weighted population variance.

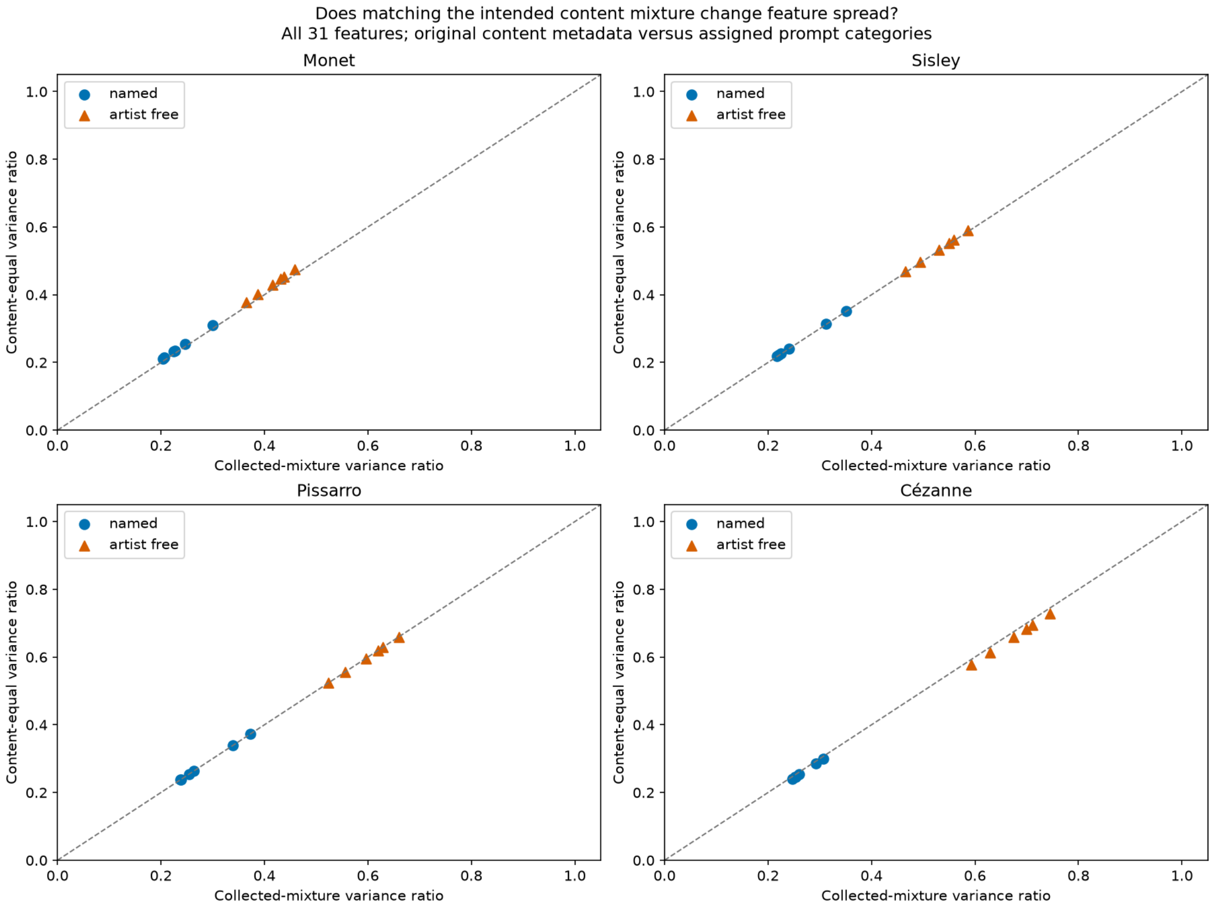

Figure 1: The fixed content-mixture sensitivity changes all-31 spread ratios little. This result concerns recorded title/prompt categories; it does not establish that visual content is adequately matched.

4.2 Comparison with finite-sample original splits

The matched-n real/real energy medians are much smaller than the corresponding generated/original medians (Table 2). This descriptive comparison argues against sample count alone explaining the observed discrepancy. It does not isolate a stylistic cause: original splits share the corpus's mixture of capture and content conditions, whereas comparisons with generated images need not.

Painter	n	Original/original	Named/original	Artist-free/original
Monet	64	.1932	2.0683	1.7319
Sisley	53	.2149	1.6776	1.8779
Pissarro	64	.1635	1.9745	1.8537
Cézanne	52	.1989	2.1419	1.6570

Table 2: All-31 matched-sample energy medians. Generated medians pool methods and aliases descriptively. Each draw uses disjoint original groups. These are finite-corpus reference distributions, not calibrated tests.

4.3 Transfer and prompt-method evidence

Cross-painter transfer yields median balanced accuracy .839 for linear and .828 for RBF models, with broad ranges .516–.952 and .500–.944. Cross-alias transfer is stronger: medians .956 and .965, with ranges .949–.973 and .947–.983. Transferability is compatible with shared generated-image or capture-related signatures. It does not imply the absence of painter-associated cues.

The earlier family-specific distance comparison found by-name prompting closest in 19 of 24 combinations of alias, painter and feature family; style-plus-aspects was closest in the other five. Explicit style instruction had larger distance than by-name prompting in all 24 comparisons. Adding aspects improved distance in 13 of 24. These observed directions motivate a narrower new intervention. They do not establish each model's best achievable prompt or justify generalizing to unobserved prompt libraries.

5 Controlled extension: protocol and technical qualification

5.1 Choice of expansion axis

We expand model-family breadth and improve the comparison target before adding more painters. New collection focuses on Monet and Cézanne. The existing four painter results are retained as development evidence. This is a disclosed two-painter case study, chosen after earlier outcomes were known; it cannot estimate the prevalence of the phenomenon across all painters or traditions.

The research grid uses the same 24 content briefs across both painters, with three repetitions, named and artist-free conditions on three routes, plus a generic named condition on OAuth: 1,008 research slots. Eighteen separate technical requests and at most 24 bounded transient-error retries give a ceiling of 1,050 attempts, including at most 612 paid attempts. The spending ceiling is $75, including the pilot and retries. Successful images are never regenerated or selected among alternatives. Sample counts reflect affordability and design coverage; they do not establish power for an image-space effect.

The reference target initially specified 48 fresh works per painter and 12 content strata. The recorded identity audit selects 64 Monet and 48 Cézanne candidates outside the prior exposure indexes. Original delivery encounters Wikimedia rate limits; a separately frozen successor follows its supported thumbnail policy. Nonupsampled standard delivery supports 38 Monet and 33 Cézanne candidates. All 71 deliveries succeed. Single-maintainer LLM visual coding excludes one figure-led Cézanne, leaving 38 Monet and 32 Cézanne works. These unequal counts are disclosed before measurement; no 48-work panel or fully balanced 12-stratum design is claimed.

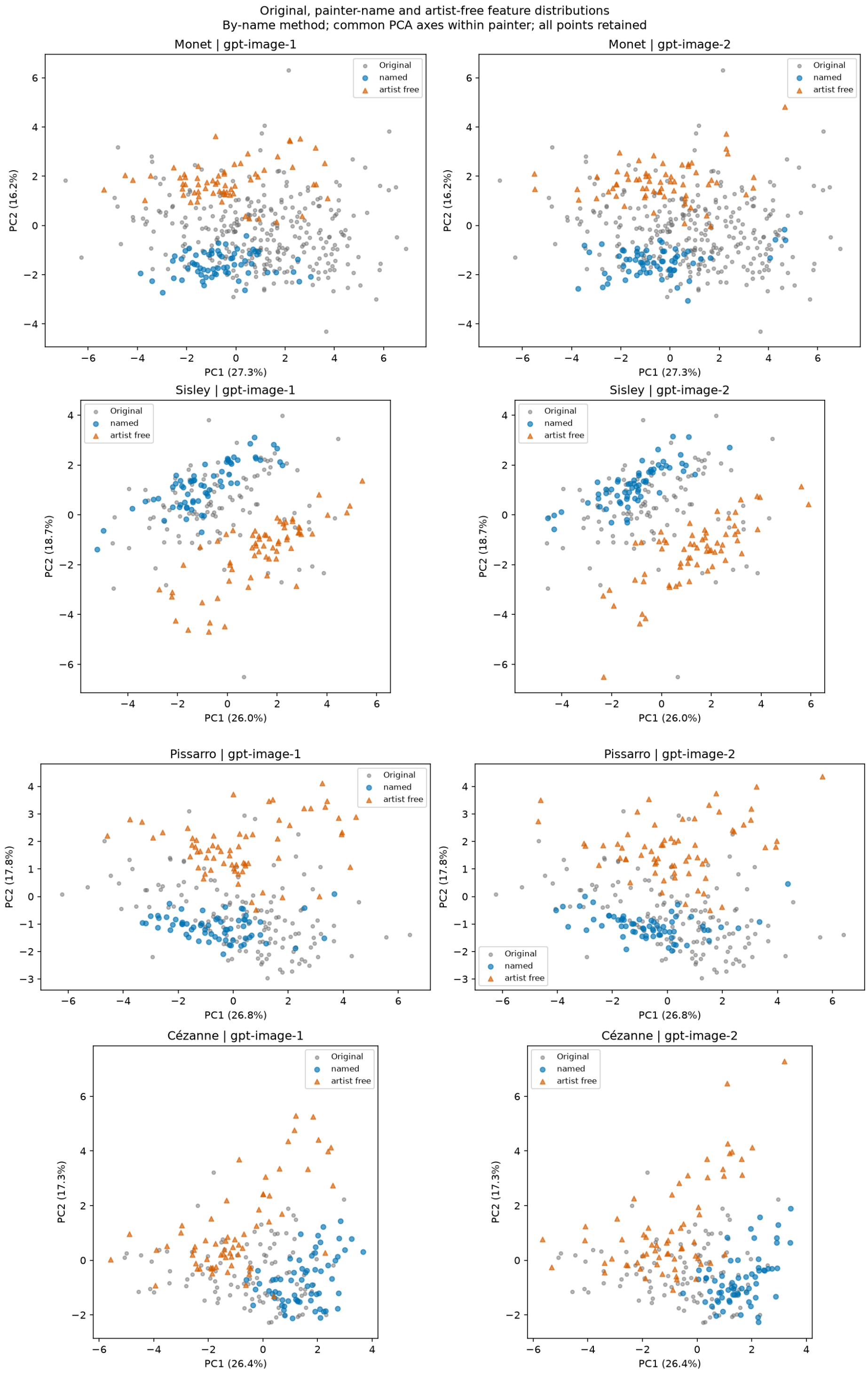

Figure 2: Named and artist-free generations in common painter-specific PCA views. Every point is retained. The unsupervised projections describe overlap and concentration; full-space group-disjoint scores provide a separate measure of detectability. The axes are not validated perceptual style dimensions.

Visually coded water, built and land organization replaces geographic-name heuristics. Counts are 21/4/13 for Monet and 3/11/18 for Cézanne. The primary comparison gives original works equal weight and generated images the same fixed class masses as their target reference. Eight shared briefs per class support this weighting. Collected-mixture and equal-class sensitivities are also required; the latter heavily weights Cézanne's three water scenes. Mixed annotations remain flagged, with an exclusion sensitivity. This is coarse content control, not exact scene matching or independent expert annotation. Independent photographic capture pairs have not been established. Thumbnail and alternate-file provenance cannot substitute for evidence of distinct capture events.

5.2 Technical pilot results

All six requests per route return decodable images. Google `gemini-3.1-flash-image` through the pinned Google AI Studio provider returns 1024-square JPEGs, costing \$0.4105 in total. BFL `flux.2-max` through its pinned provider returns 1024-square PNGs, costing \$0.4200. The local OAuth `gpt-image-2` comparator returns nonsquare PNGs with short sides 1121–1126 and reports low rather than requested medium quality. No paid fallback, retry, prompt rewrite or feature-based pilot selection occurs.

The total paid pilot cost is \$0.8304855. Using each paid route's maximum observed cost projects the remaining 576 paid requests to \$39.8917. Adding 25% headroom and the pilot gives \$50.6952. Including all 24 potential retries at the highest observed unit cost, with the same headroom, gives \$52.7952 within the \$75 ceiling. This projection is not a guarantee of future pricing. The worker reserves \$5 before a paid request and stops when an outcome or cost falls outside the declared accounting rules, or the spending guard would be crossed.

Both paid routes share observed geometry, but their encodings differ. OAuth is a separate service comparison where matched rendering is required. Symmetric 256-short-side and common-JPEG sensitivities, with development-only scaling, must accompany the native-processing comparison. Such sensitivities cannot undo unknown prior compression or demonstrate equivalent physical capture.

5.3 Controlled measurement and distribution endpoints

The controlled reference panel has now produced all 210 expected feature vectors under three processing pipelines, without measurement failures or duplicate raw image hashes. Recomputing those vectors from retained source bytes reproduces all records. Both sensitivity scalers were fitted first, from 442 remeasurements of the same 221 historical development works. Research generation remains in progress; generated evaluation features are withheld until its terminal receipt.

For painter p, let N_{pc} denote the number of reference works in content class c, N_p the total, and n_{pc} the number of available generated observations in a particular route/condition cell. The primary weights are

$$a_i = 1/N_p, \qquad b_j = \frac{N_{pc}}{N_p n_{pc}} \quad \text{for } j \in c. \tag{2}$$

Each cell is assigned 72 slots: eight briefs per content class, each repeated three times. Missing outputs redistribute only their own class's fixed mass; a required class with no measured output makes that comparison unavailable. The weighting is conditional on intended generated content, whose adherence has not been independently judged. Images are not discarded for failing to resemble a painter or for depicting the intended scene imperfectly. Effective weight size $1/\sum_j b_j^2$ is reported separately from the number of image files.

Primary energy discrepancy and the two spread ratios use all 31 standardized features. Color, spatial and texture subsets remain descriptive. Prespecified sensitivities use the collected generated mixture, one-third mass per content class, only unambiguous reference annotations, only reference sources with native short sides at least 1024 pixels, and initial-attempt successes alone. The annotation and native-size restrictions redefine their reference class masses from the retained subset and are labeled accordingly. A 256-pixel short side and common JPEG encoding at quality 90 with 4:2:0 subsampling additionally probe processing sensitivity. No histogram, palette or texture matching is performed.

For the new disjoint original/original baseline, each of 999 deterministic draws selects two nonoverlapping sets of $\lfloor N_{pc}/2 \rfloor$ works per class and a generated set with the same class counts, without replacement. All sets retain the declared class masses. The resulting group sizes are 18 for Monet and 15 for Cézanne; the latter includes only one water work per group. The repeated draws characterize this finite panel and do not create independent reference samples.

Secondary coverage uses an original-centered ball with radius r_i equal to the distance to the third nearest distinct reference work:

$$\widehat{C} = \sum_i a_i \, \mathbf{1}\left\{ \min_j \|x_i - y_j\|_2 \le r_i \right\}. \tag{3}$$

We report all-available coverage and 100 uniform generated subsamples of size $\min(N_p, n_p)$, without replacement. These neighborhood measurements depend on sample size and representation and are not artistic-completeness scores.

Fixed linear/RBF kernel-ridge detection uses six whole-brief folds and disjoint reference works, with no tuning or PCA input. Reference folds are assigned by content-stratified identity hashes, so sparse classes are distributed across folds. Training assigns equal total mass to original and generated labels and matches content masses within each label; held-out balanced accuracy and AUC use the same target masses. Eight folds over assigned collection batches provide a timing sensitivity; the batches do not identify independent service sessions. Every split and prediction is retained. For visualization, each painter has one common PCA fit with half the mass on originals and half shared equally across seven generated cells, using the primary content weights. Original-only PCA is also retained. Every point is shown under shared within-painter axes. Painter specificity compares each named cell with both reference painters under the same one-third-per-class mixture, separately from its absolute discrepancy.

5.4 Prospective inference and missingness boundary

Research generation requires a committed reference, prompt, endpoint, schedule and analysis freeze before evaluation features are extracted. The original design assigned repetitions of each brief to distinct batches among eight planned windows with offsets 0, 3, 6, 9, 24, 27, 30 and 33 hours. An execution amendment closed the unused sequential collector before its first research request and specified three route workers, at most one active request per route, with globally staggered starts at least five seconds apart. Within-route randomized condition order is preserved. Actual timestamps, attempted slots, refusals and uncertain requests are retained. Scheduling across days does not by itself establish independent sessions; concurrent gateway traffic also leaves the no-interference assumption unverified.

The first parallel census closed after 48 attempted slots: 47 returned images and one explicit OAuth moderation refusal encoded as HTTP 400. A generic status rule had classified that refusal as a request-contract error. A prospective correction recognizes the verified, zero-charge moderation envelope as a failed observation and continues only the 960 unattempted slots, retaining predecessor failure history for the cluster stop. The blocked slot is not retried, rewritten or rerouted. The final derived view will retain both censuses, their interruption and every original slot identity. This correction preceded all generated-feature measurement; the 47 existing images were not selected by their measured outcomes.

The continuation subsequently returned 51 further images and one complete FLUX 502 submission error without image data, a generation ID or reported cost. OpenRouter documents a billing waiver for failed image calls (OpenRouter, 2026). Because that policy is not an individual billing receipt, the recovery keeps the missing cost unchanged and reserves its full \$5 against the study ceiling. Only a hash-verified, complete error-only response meeting the declared transient-error contract receives a single delayed retry. Incomplete or unclassified outcomes still stop collection. The recovery returned 404 further initial images and one successful, identical-payload retry of the FLUX failure. It retained all 98 prior images and both initial failures, without retrying the OAuth moderation refusal.

After the first four assigned batches, the user requested immediate completion without the remaining night waits. This change preceded all generated-feature measurement. The waiting recovery was closed with no unresolved request, and a fourth census retains the 503 available images while collecting the 504 untouched slots in consecutive batches. Within-route order, prompt assignments, rate limits, references, scalers and statistical endpoints are unchanged; a five-second pause separates drained batches. Original planned times and batch IDs remain in the record alongside actual timestamps. The original 33-hour schedule was therefore not executed in full, and the last four batches do not establish independent days or backend sessions.

Prospective synthetic qualification passes for the eight paired all-31 prompt contrasts: six named-versus-free and two detailed-versus-generic OAuth contrasts. Two fixed-seed phases each use 10,000 trials in 24 null cells covering pair counts, contribution heterogeneity, window drift and endpoint dependence. Actual tests use 99,999 Monte Carlo sign draws and Holm adjustment. Their sharp null concerns availability and features under the randomized slot policy, conditional on the finite reference, with no interference. Service carryover and retry-induced timing remain limitations. This qualification does not validate absolute-distribution p-values, oeuvre-sampling intervals or actual image-space power. Absolute gaps and leave-batch/brief ranges remain descriptive. Ordinary image-level bootstrap intervals are not justified by the number of files alone. No significance-based sampling extension, silent replacement of missing outputs or retrospective change to a sealed endpoint is allowed. No new-model fidelity result, p-value or human result is available in this prototype revision.

5.5 Paired statistic and conditional interpretation

For a matched before/after pair (A_i, B_i) with fixed class-adjusted weight b_i, define $C(v) = \sum_j a_j \|v - x_j\|_2$ and $T(v) = \sum_i b_i(\|v - A_i\|_2 + \|v - B_i\|_2)$. The contribution

$$c_i = b_i\{2[C(B_i) - C(A_i)] - [T(B_i) - T(A_i)]\} \tag{4}$$

satisfies $\sum_i c_i = \widehat{\mathcal{E}}(X, B) - \widehat{\mathcal{E}}(X, A)$. Thus the contrast retains the within-generated pairwise terms; it is not a contrast of mean-to-centroid distances. Complementary condition swaps produce $\sum_i s_i c_i$ for assignment signs $s_i \in \{-1, 1\}$. For the three-condition OAuth groups, each pairwise comparison conditions on the third condition's position in the uniformly randomized order.

A contrast uses complete measured pairs sharing a brief, repetition and assigned batch. If fewer than 24 pairs or 12 distinct briefs survive, or a required content class is absent, its endpoint is unavailable and retains $p = 1$ in the fixed eight-test family. This is distinct from the all-available descriptive comparison. Two-sided Monte Carlo p-values use $(1+e)/(1+99999)$, where e counts sign-draw statistics at least as extreme in absolute value, including numerical ties. Leave-one-batch and leave-one-brief estimates assess sensitivity without being labeled confidence intervals. The sharp assignment null covers both availability and measured features for the full fixed slot policy, including its bounded technical retry rule; it is stronger than equality of expected distances. Treatment-dependent service duration and retry delays can affect later calls, so the randomization calculation does not remove the no-interference limitation.

6 Discussion

The completed results support the narrow empirical hypothesis of a feature-space gap and reduced spread in the tested generated collections. They provide weaker support for the stronger claim that generation intrinsically fails to reproduce a painter's style distribution. In particular, artist-free detection and cross-alias transfer suggest a component shared across generation conditions; coarse content weighting does not eliminate finer composition differences; and digital reproduction remains unresolved.

A productive interpretation separates recognition from distribution reproduction. Painter naming may improve proximity or specificity without recovering the range of original works. Conversely, a strong detector may respond to properties that matter little to human observers. The present features cannot establish recognizability, authenticity, copying, aesthetic merit or equivalence to an artist's practice. Those claims require additional construct validation.

The next empirical results should therefore be judged by how controls change effect patterns. Persistence across new families and stronger content controls would broaden the evidence for a distributional gap. A substantial reduction after content or processing controls would support those explanations instead. Either outcome is scientifically useful. Selecting the interpretation only after seeing which result favors the initial hypothesis would weaken the contribution.

7 Limitations and reproducibility

The original corpus is a finite collection of available digital surrogates, not a probability sample of each painter's oeuvre. The new reference panel is small and availability constrained. Service aliases and live model slugs lack attested immutable weight snapshots. Prompt repetition, shared briefs and service state can induce dependence. Title-based development labels and later maintainer-LLM visual annotations are not independent expert judgments. All review so far is maintainer/LLM self-review; none is institutionally independent.

The paper source, compact manifests, exact prompts, preprocessing code, report tables, split memberships and hashes are retained in the repository. Raw artwork and complete generated-image responses remain in an ignored local evidence workspace; a Git checkout alone does not reconstruct those bytes. Acquisition licenses and source identities are recorded, and failed responses remain evidence. The new study uses its own namespace and does not modify historical terminal runs.

Human evaluation is a prepared follow-up, not an executed component. Expert review could assess painter recognizability and perceived variation using blinded, balanced stimuli and independent annotation. Learned representations are also deferred; adding a learned evaluator would not automatically establish that its distance is calibrated to human style judgments. Neither a collaborator's participation nor authorship is presumed in this prototype.

A Reproduction entry points

From the repository root, the completed numeric development bundle can be checked without provider calls or feature extraction:

```
uv run --locked --extra analysis --extra learned python -m
latent_art_bench.painter_distribution_study_v1.diagnostics check
```

The technical pilot's costs, container metadata and retained raw responses are checked with:

```
uv run --locked python -m
latent_art_bench.painter_distribution_study_v1.pilot_summary check
```

The optional environment label `learned` in the first command supplies existing dependencies; this analysis does not download or run a learned encoder. Canonical stage status is in `docs/STATUS.md`. Frozen protocols and terminal ledgers must not be edited to make an audit pass. Exact software dependencies are pinned by `uv.lock`. Raw image acquisition and generation commands are separate from the standard offline test suite.

References

Andrea Asperti. On the separation of human and AI-generated images in CLIP embedding space. arXiv:2608.25609, version 1, 2026. URL https://arxiv.org/abs/2608.25609. Preprint.

D. Kim, S.-W. Son, and H. Jeong. Large-scale quantitative analysis of painting arts. *Scientific Reports*, 4:7370, 2014. doi: 10.1038/srep07370.

J. Kim, B. Lee, T. You, and J. Yun. Context-aware multimodal AI navigates hidden pathways in five centuries of art evolution. *Proceedings of the National Academy of Sciences*, 123(30): e2517969123, 2026. doi: 10.1073/pnas.2517969123.

B. Lee, M. K. Seo, D. Kim, I.-S. Shin, M. Schich, H. Jeong, and S. K. Han. Dissecting landscape art history with information theory. *Proceedings of the National Academy of Sciences*, 117 (43):26580–26590, 2020. doi: 10.1073/pnas.2011927117.

Muhammad Ferjad Naeem, Seong Joon Oh, Youngjung Uh, Yunjey Choi, and Jaejun Yoo. Reliable fidelity and diversity metrics for generative models. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 7176–7185, 2020. URL https://proceedings.mlr.press/v119/naeem20a.html.

OpenRouter. Image generation models on OpenRouter, July 27 2026. URL https://openrouter.ai/blog/tutorials/image-generation-models/. Provider documentation; accessed 6 September 2026.

Gaurav Parmar, Richard Zhang, and Jun-Yan Zhu. On aliased resizing and surprising subtleties in GAN evaluation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. URL https://arxiv.org/abs/2104.11222.

David R. Roberts, Volker Bahn, Simone Ciuti, et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017. doi: 10.1111/ecog.02881.

H. Y. D. Sigaki, M. Perc, and H. V. Ribeiro. History of art paintings through the lens of entropy and complexity. *Proceedings of the National Academy of Sciences*, 115(37):E8585–E8594, 2018. doi: 10.1073/pnas.1800083115.

Gábor J. Székely and Maria L. Rizzo. Energy statistics: A class of statistics based on distances. *Journal of Statistical Planning and Inference*, 143(8):1249–1272, 2013. doi: 10.1016/j.jspi.2013.03.018.